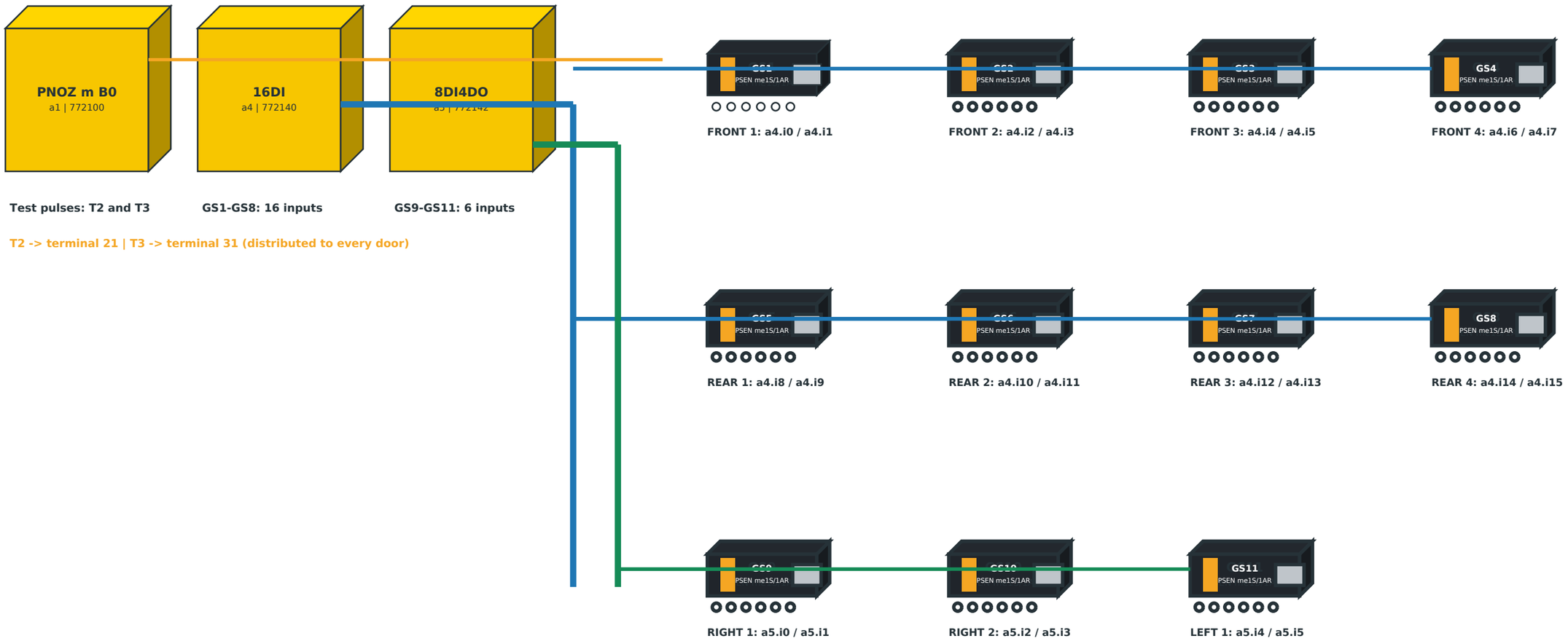
11 DOOR SAFETY ARCHITECTURE - 3D OVERVIEW

Confirmed PNOZ input allocation and PSEN terminal functions. Unlock output source remains a release hold point.



Test pulses: T2 and T3

GS1-GS8: 16 inputs

GS9-GS11: 6 inputs

T2 -> terminal 21 | T3 -> terminal 31 (distributed to every door)

FRONT 1: a4.i0 / a4.i1

FRONT 2: a4.i2 / a4.i3

FRONT 3: a4.i4 / a4.i5

FRONT 4: a4.i6 / a4.i7

REAR 1: a4.i8 / a4.i9

REAR 2: a4.i10 / a4.i11

REAR 3: a4.i12 / a4.i13

REAR 4: a4.i14 / a4.i15

RIGHT 1: a5.i0 / a5.i1

RIGHT 2: a5.i2 / a5.i3

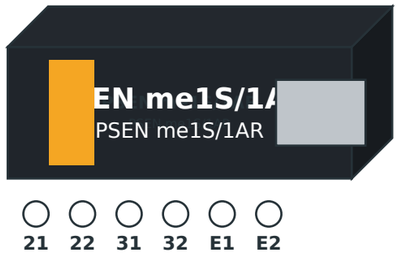
LEFT 1: a5.i4 / a5.i5

SAFE UNLOCK BUS: SAFE_UL_24V -> local fuse F-ULx -> relay/contact KGSx -> E1; E2 -> 0 V

HOLD POINT: exact PNOZ safe output terminal and cabinet terminal numbers are not yet proven - release from verified project/electrical drawing.

ONE DOOR - DETAILED SENSOR AND 8-CORE CABLE

Repeat this circuit for GS1 through GS11; only the assigned CH1 and CH2 PNOZ inputs change.

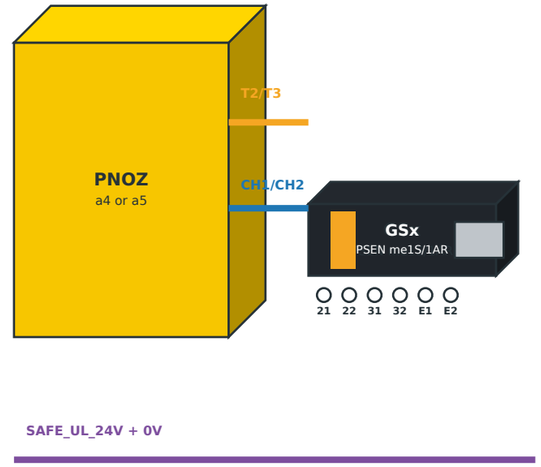


Sensor terminals

21-22	NC guard-position safety contact
31-32	NC lock-position safety contact
E1-E2	24 V unlock solenoid
13-14	Auxiliary NO contact (optional)
43-44	Auxiliary NO contact (optional)

Recommended cable: 8 conductors per door

Core 1: T2	-> PSEN 21
Core 2: PSEN 22	-> PNOZ CH1 input
Core 3: T3	-> PSEN 31
Core 4: PSEN 32	-> PNOZ CH2 input
Core 5: SAFE_UL_24V via F-ULx/KGSx	-> PSEN E1
Core 6: PSEN E2	-> 0 V DC
Core 7: Spare / optional auxiliary	-> 13-14 or 43-44
Core 8: Spare / optional return	-> non-safety input/common



Cable count rule

Minimum functional: 6 conductors

Recommended: 8 conductors (2 spare/aux)

Both auxiliary contacts used: 10 conductors

Cable cross-section, wire numbers, fuses and cabinet terminal blocks must be released by the electrical design; they are intentionally not invented here.

COMPLETE DOOR SCHEDULE AND INSTALLATION TOTALS

Exact confirmed input assignment from the PNOZmulti project screenshots supplied by the operator.

Door	Location	PSEN safety path	CH1 Pilz input	CH2 Pilz input	Cable	Unlock branch
GS1	FRONT 1	T2-21/22 + T3-31/32	a4.i0	a4.i1	8-core	F-ULx / KGSx -> E1-E2
GS2	FRONT 2	T2-21/22 + T3-31/32	a4.i2	a4.i3	8-core	F-ULx / KGSx -> E1-E2
GS3	FRONT 3	T2-21/22 + T3-31/32	a4.i4	a4.i5	8-core	F-ULx / KGSx -> E1-E2
GS4	FRONT 4	T2-21/22 + T3-31/32	a4.i6	a4.i7	8-core	F-ULx / KGSx -> E1-E2
GS5	REAR 1	T2-21/22 + T3-31/32	a4.i8	a4.i9	8-core	F-ULx / KGSx -> E1-E2
GS6	REAR 2	T2-21/22 + T3-31/32	a4.i10	a4.i11	8-core	F-ULx / KGSx -> E1-E2
GS7	REAR 3	T2-21/22 + T3-31/32	a4.i12	a4.i13	8-core	F-ULx / KGSx -> E1-E2
GS8	REAR 4	T2-21/22 + T3-31/32	a4.i14	a4.i15	8-core	F-ULx / KGSx -> E1-E2
GS9	RIGHT 1	T2-21/22 + T3-31/32	a5.i0	a5.i1	8-core	F-ULx / KGSx -> E1-E2
GS10	RIGHT 2	T2-21/22 + T3-31/32	a5.i2	a5.i3	8-core	F-ULx / KGSx -> E1-E2
GS11	LEFT 1	T2-21/22 + T3-31/32	a5.i4	a5.i5	8-core	F-ULx / KGSx -> E1-E2

11 x 8-core
door cables

88
installed conductors

66
essential conductors used

22
spare / optional auxiliary

Commissioning check: verify each physical door identity, both channel transitions, cross-short diagnostics, lock state, controlled unlock and fault reset.

Do not energize or bypass the safety function from this concept drawing. Use the released schematic and validated PNOZ project for final commissioning.

EVERY WIRE FROM PILZ TO EVERY DOOR SENSOR

Six individual conductors shown for each PSEN me1S/1AR. CH1 and CH2 are the confirmed PNOZ inputs.

PNOZmulti 2

B0 test pulses + a4/a5 inputs

safe unlock distribution

- T2 -> 21
- 22 -> CH1
- T3 -> 31
- 32 -> CH2
- SAFE_UL_24V -> E1
- E2 -> 0V

GS1 CH1=a4.i0 CH2=a4.i1

T2 -> 21
22 -> a4.i0
T3 -> 31
32 -> a4.i1
SAFE_UL_24V -> E1
E2 -> 0V

GS1
PSEN me1S/1AR
21 22 31 32 E1 E2

GS2 CH1=a4.i2 CH2=a4.i3

T2 -> 21
22 -> a4.i2
T3 -> 31
32 -> a4.i3
SAFE_UL_24V -> E1
E2 -> 0V

GS2
PSEN me1S/1AR
21 22 31 32 E1 E2

GS3 CH1=a4.i4 CH2=a4.i5

T2 -> 21
22 -> a4.i4
T3 -> 31
32 -> a4.i5
SAFE_UL_24V -> E1
E2 -> 0V

GS3
PSEN me1S/1AR
21 22 31 32 E1 E2

GS4 CH1=a4.i6 CH2=a4.i7

T2 -> 21
22 -> a4.i6
T3 -> 31
32 -> a4.i7
SAFE_UL_24V -> E1
E2 -> 0V

GS4
PSEN me1S/1AR
21 22 31 32 E1 E2

GS5 CH1=a4.i8 CH2=a4.i9

T2 -> 21
22 -> a4.i8
T3 -> 31
32 -> a4.i9
SAFE_UL_24V -> E1
E2 -> 0V

GS5
PSEN me1S/1AR
21 22 31 32 E1 E2

GS6 CH1=a4.i10 CH2=a4.i11

T2 -> 21
22 -> a4.i10
T3 -> 31
32 -> a4.i11
SAFE_UL_24V -> E1
E2 -> 0V

GS6
PSEN me1S/1AR
21 22 31 32 E1 E2

GS7 CH1=a4.i12 CH2=a4.i13

T2 -> 21
22 -> a4.i12
T3 -> 31
32 -> a4.i13
SAFE_UL_24V -> E1
E2 -> 0V

GS7
PSEN me1S/1AR
21 22 31 32 E1 E2

GS8 CH1=a4.i14 CH2=a4.i15

T2 -> 21
22 -> a4.i14
T3 -> 31
32 -> a4.i15
SAFE_UL_24V -> E1
E2 -> 0V

GS8
PSEN me1S/1AR
21 22 31 32 E1 E2

GS9 CH1=a5.i0 CH2=a5.i1

T2 -> 21
22 -> a5.i0
T3 -> 31
32 -> a5.i1
SAFE_UL_24V -> E1
E2 -> 0V

GS9
PSEN me1S/1AR
21 22 31 32 E1 E2

GS10 CH1=a5.i2 CH2=a5.i3

T2 -> 21
22 -> a5.i2
T3 -> 31
32 -> a5.i3
SAFE_UL_24V -> E1
E2 -> 0V

GS10
PSEN me1S/1AR
21 22 31 32 E1 E2

GS11 CH1=a5.i4 CH2=a5.i5

T2 -> 21
22 -> a5.i4
T3 -> 31
32 -> a5.i5
SAFE_UL_24V -> E1
E2 -> 0V

GS11
PSEN me1S/1AR
21 22 31 32 E1 E2

Per door cable

6 cores used

8-core recommended

2 spare/auxiliary